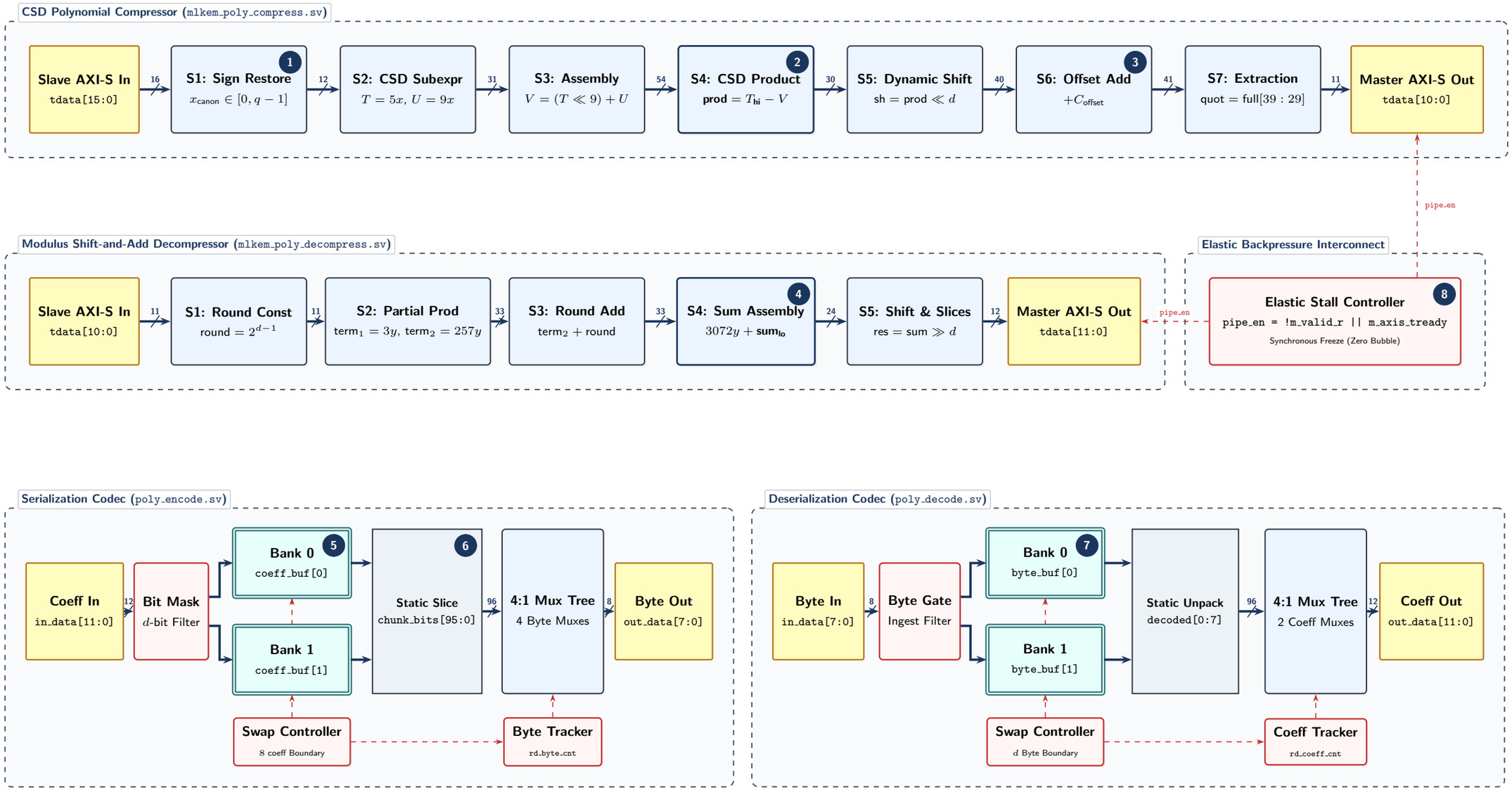
FIPS 203 ML-KEM Architecture: Multiplier-Free Polynomial Codecs & Zero-Bubble Serialization Subsystem

Subsystem 10 Microarchitecture — RTL Modules: `mlkem_poly_compress.sv`, `mlkem_poly_decompress.sv`, `poly_encode.sv`, `poly_decode.sv`



ID	Component / Signal	Width / Format	Description
1	Input Reg & Signed Restore	16b → 12b	Maps signed coefficients $[-q, q-1]$ into canonical range $[0, q-1]$ via $x[15]\,?\,(x+3329) : x$.
2	Multiplier-Free CSD Tree	12b → 30b	Evaluates $161271 \cdot x = (5x \ll 15) - (2560x + 9x)$ using 4 shift-add operations without DSPs.
3	Dynamic Shift & Offset Adder	30b → 11b	Computes $(\text{prod} \ll d) + C_{\text{offset}}$ ($C_{\text{offset}} = 268354944$) and slices bits $[39 : 29]$ for FIPS 203 rounding.
4	Shift-and-Add Decompressor	11b → 12b	Reconstructs $(3329y + 2^{d-1}) \gg d$ using 5-stage shift-add decomposition $3329 = 3072 + 257$ with 0 DSPs.
5	Dual-Bank Ping-Pong Buffer	$2 \times (8 \times 12\text{b})$	Two banks of eight 12-bit registers alternating write/read on 8-coefficient chunk boundaries ($8d\text{b} = d\text{B}$).
6	Static Bit-Slice & 4:1 Mux	96b → 8b	Concatenates 8 coefficients into 96-bit bus and serializes bytes via hierarchical 4:1 mux tree without shifters.
7	Deserialization Ping-Pong Array	$2 \times (12 \times 8\text{b})$	Dual-bank byte buffer unpacking 96-bit chunks into 8 parallel coefficients, serialized by hierarchical mux tree.
8	Elastic Stall Controller	Control	Propagates downstream ready/valid handshakes and freezes internal registers synchronously under stall.